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Digital Image Processing

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CONTENTS

IMAGE RESTORATION

Image Degradation and Restoration

- When an image f(x, y) is degraded (processed into less quality) into a degraded image g(x, y) by operation $\mathcal{H}$ plus noise η. By having some knowledge about $\mathcal{H}$ and η, it might be possible to restore the estimation of original image $\hat{f}(x, y)$

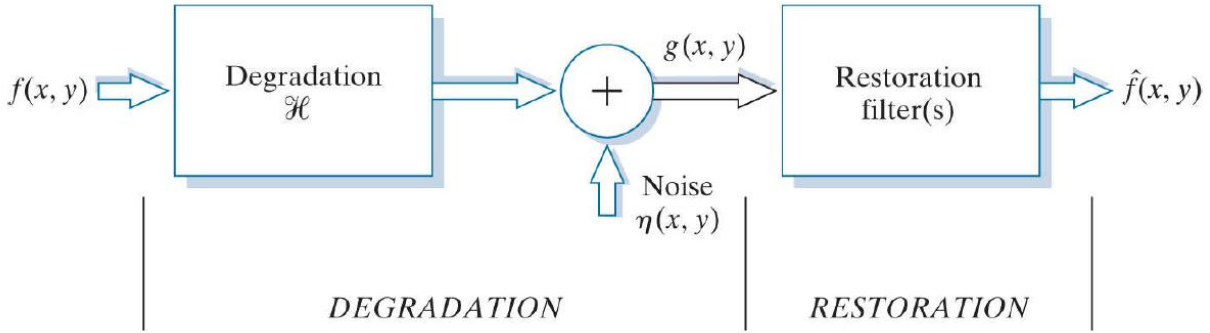


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Image Degradation and Restoration

- f(x, y) – image before degradation, ‘true image’
- g(x, y) – image after degradation, ‘observed image’
- h(x, y) – degradation filter
- $\hat{f}(x, y)$ – estimate of f(x,y) computed from g(x,y)
- η(x, y) – additive noise

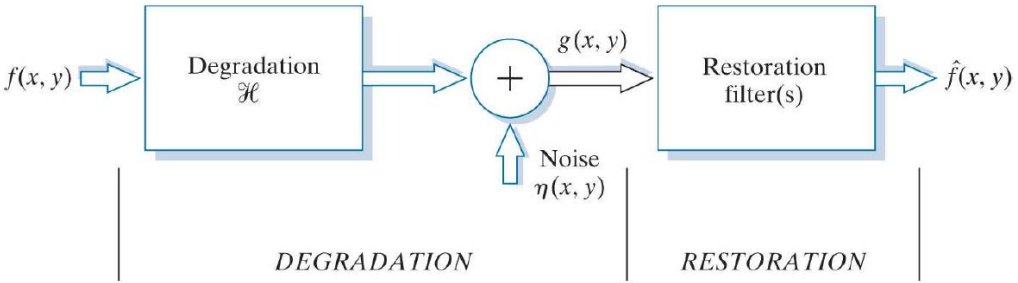


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Image Degradation and Restoration

• original

• optical blur

• motion blur

• spatial quantization (discrete pixels)

• additive intensity noise

Image from: A. Zisserman (2014). B14 Image Analysis, University of Oxford

Image Degradation and Restoration

- If $\mathcal{H}$ is a linear, position-invariant operator, the degraded image in the spatial domain is

$$g(x,y) = (h \star f)(x,y) + \eta(x,y)$$

- Where $\star$ indicated convolution and h(x, y) is the spatial representation of the degradation function
- On the other hand, the degraded image in the frequency domain is

$$G(u,v) = H(u,v)F(u,v) + N(u,v)$$

Noise Reduction

Noise Models

- The principal sources of noise in digital images arise during image acquisition and/or transmission
- The performance of imaging sensors is affected by a variety of environmental factors during image acquisition, and by the quality of the sensing elements themselves

Probability Density Function

- Probability Density Function (PDF) is a function of each noise model which indicates the chance of noise within an image. The PDF of some noise models are as below

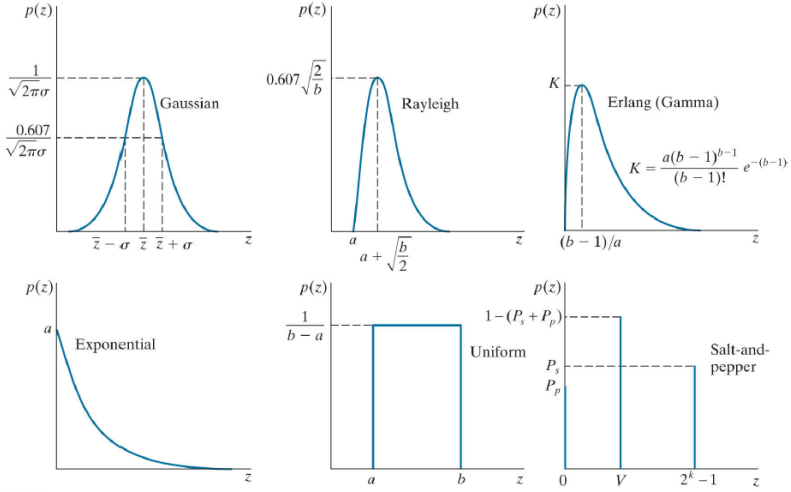


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersle

Noisy images and their histograms

- We will now test the image below with different models of noise and compare their histograms and their PDF function

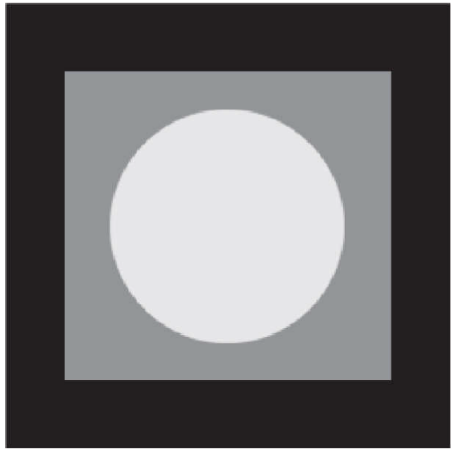
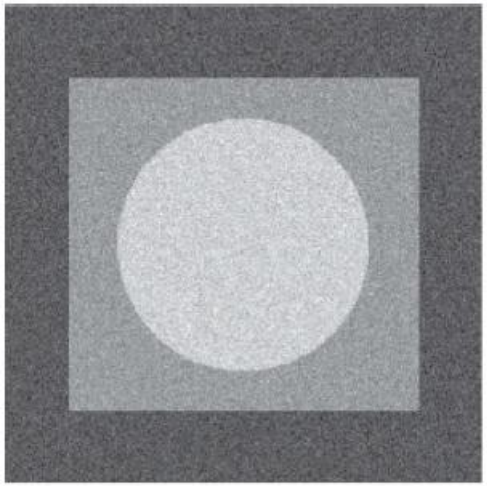
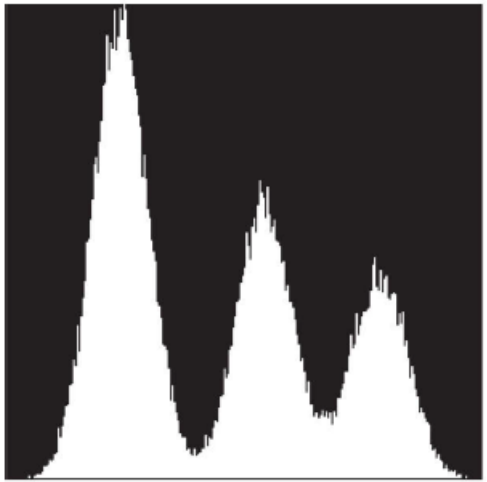


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Noisy images and their histograms

- Gaussian noise





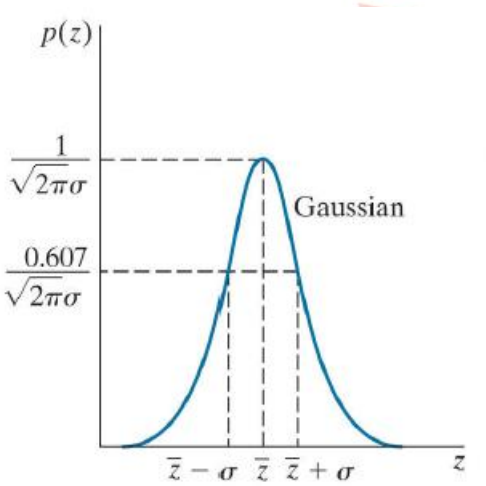
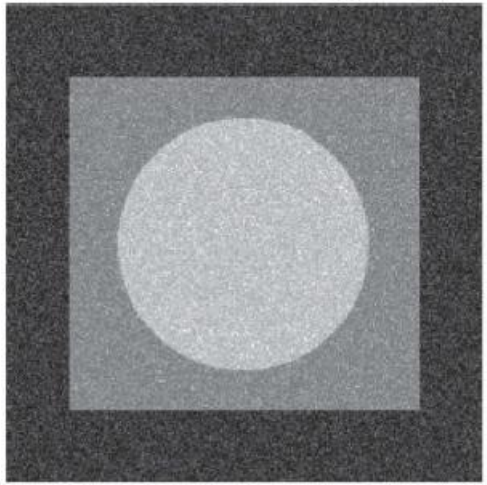
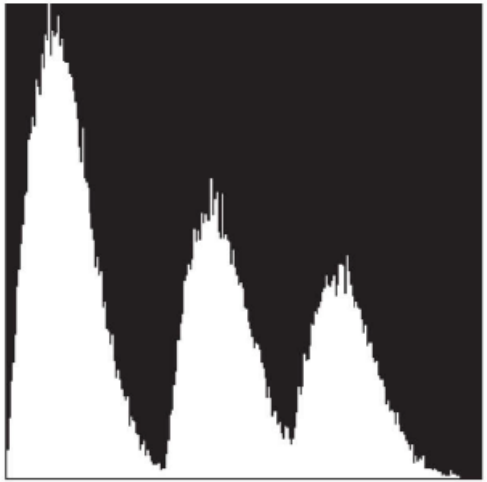


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Noisy images and their histograms

- Rayleigh noise





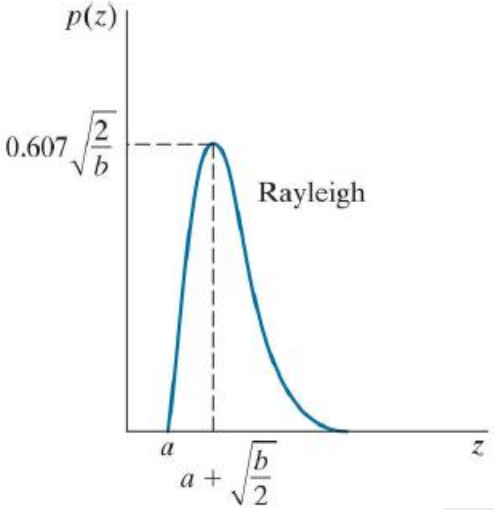
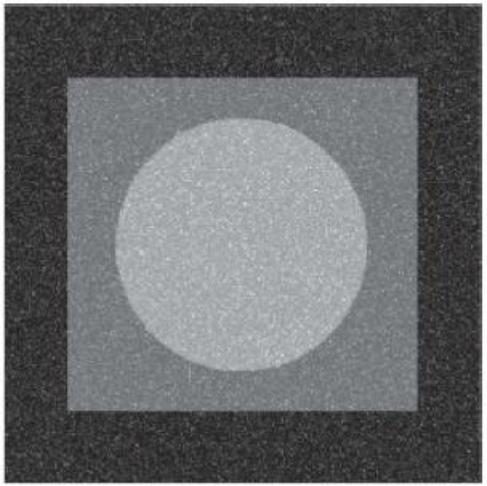


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Noisy images and their histograms

- Erlanga noise



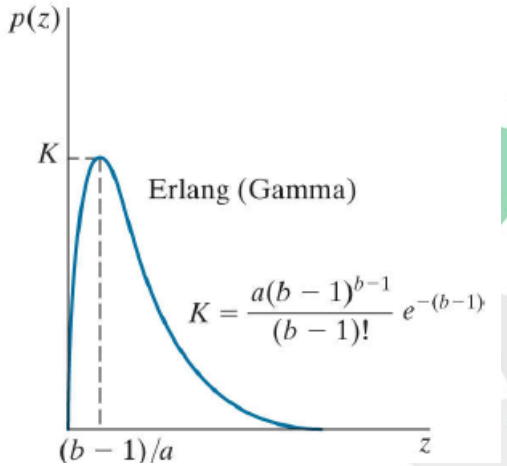
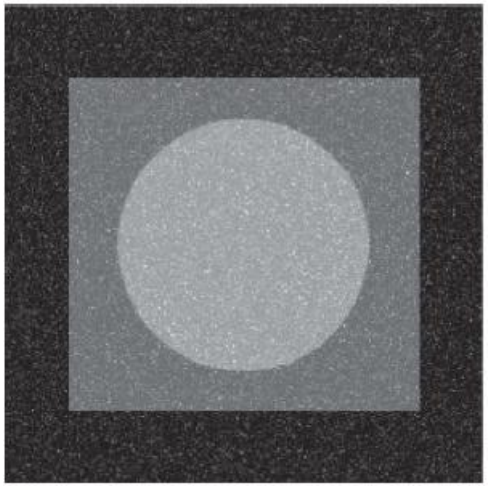


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Noisy images and their histograms

- Exponential noise



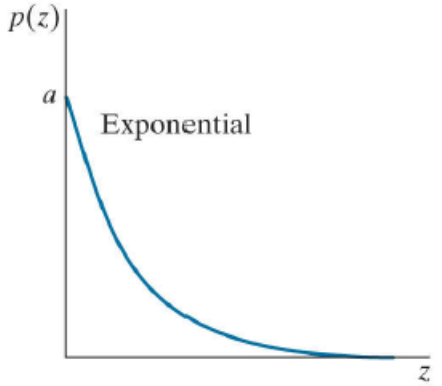
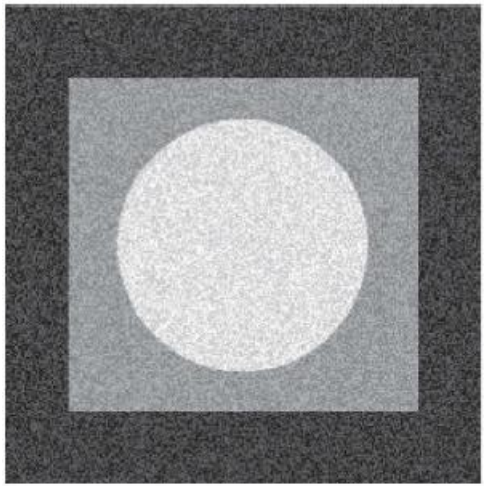
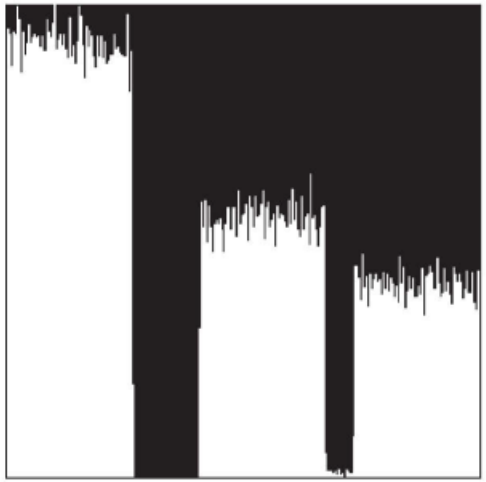


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Noisy images and their histograms

- Uniform noise





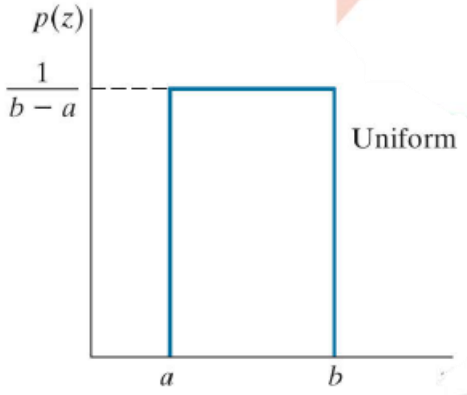
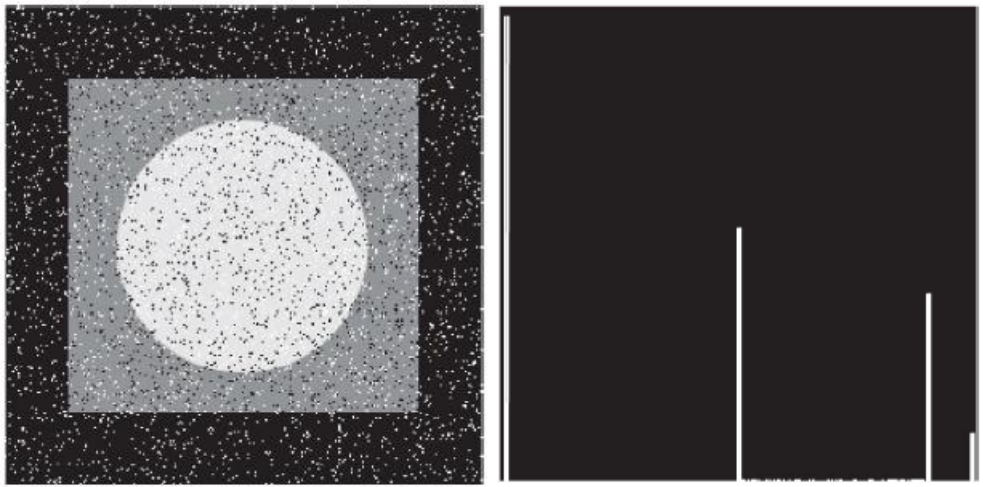


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Noisy images and their histograms

- Salt-and-pepper noise



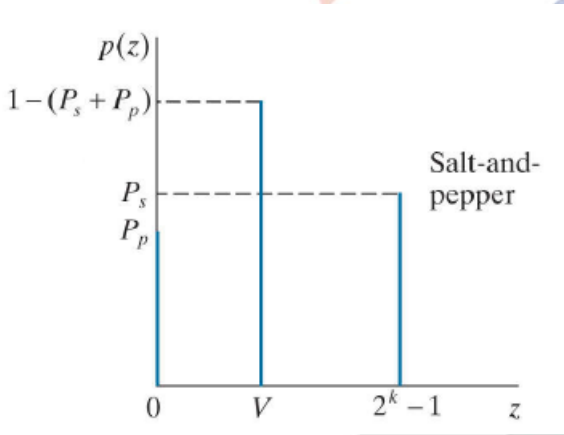


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Periodic Noise

- Periodic noise typically arises from electrical or electromechanical interference during image acquisition.

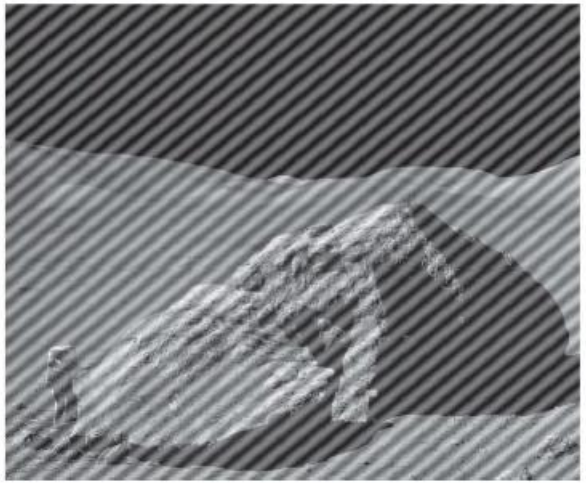


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Noise Reduction – Spatial Filtering

Denoising image

- If we consider only the presence of noise, an image is degraded as below

$$g(x,y) = f(x,y) + \eta(x,y)$$

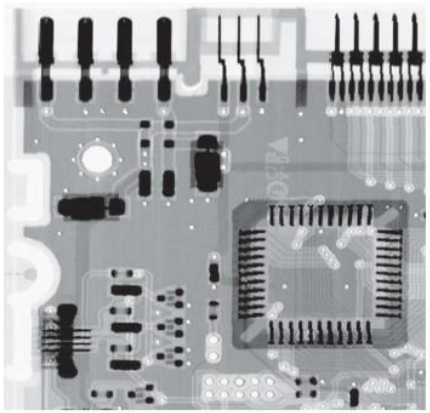
$$G(u,v) = F(u,v) + N(u,v)$$

- In order to denoise, we can use spatial filtering when only random noise is present

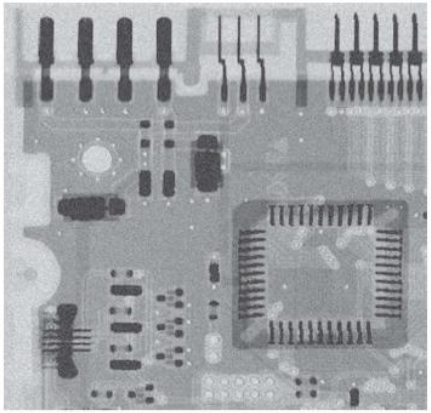
Mean Filters - Arithmetic

Simplest Mean Filter

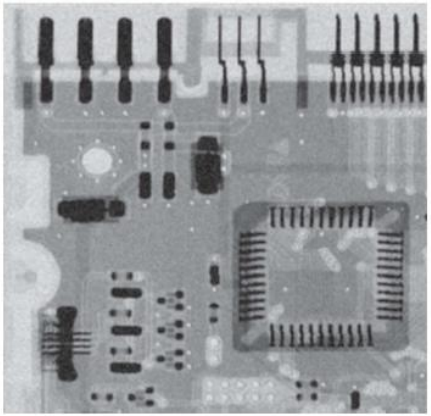
$$\hat{f}(x,y) = \frac{1}{mn} \sum_{(r,c) \in S_{xy}} g(r,c)$$



Original Image



Corrupted with Gaussian noise



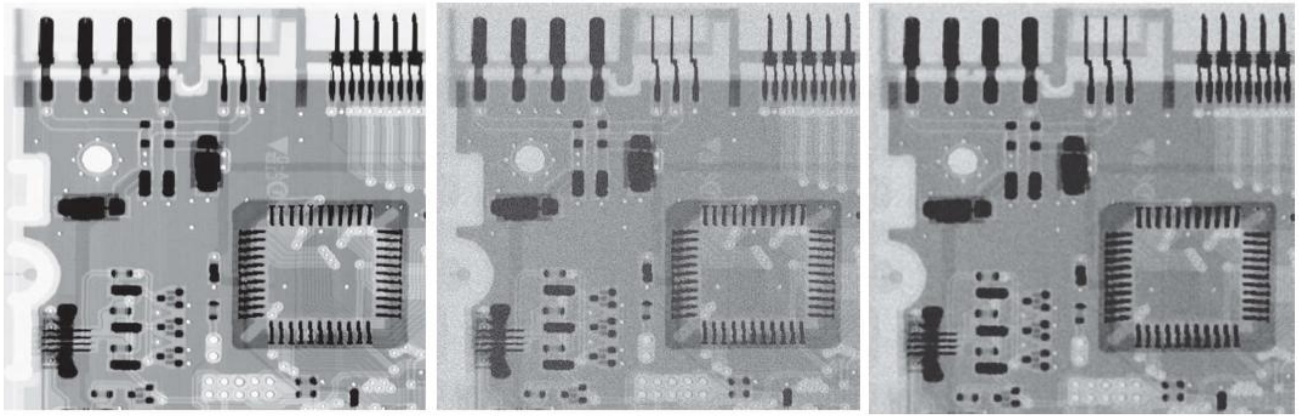
Filtered with Arithmetic Mean Filter

Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Mean Filters - Geometric

Same level of smoothing result as arithmetic, better in losing detail

$$\hat{f}(x,y) = \left[\prod_{(r,c) \in S_{xy}} g(r,c) \right]^{\frac{1}{mn}}$$



Original Image

Corrupted with Gaussian noise

Filtered with Geometric Mean Filter

Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Mean Filters - Harmonic

Works well for salt noise, fails for pepper noise.

$$\hat{f}(x,y) = \frac{mn}{\sum_{(r,c) \in S_{xy}} \frac{1}{g(r,c)}}$$

Mean Filters - Contraharmonic

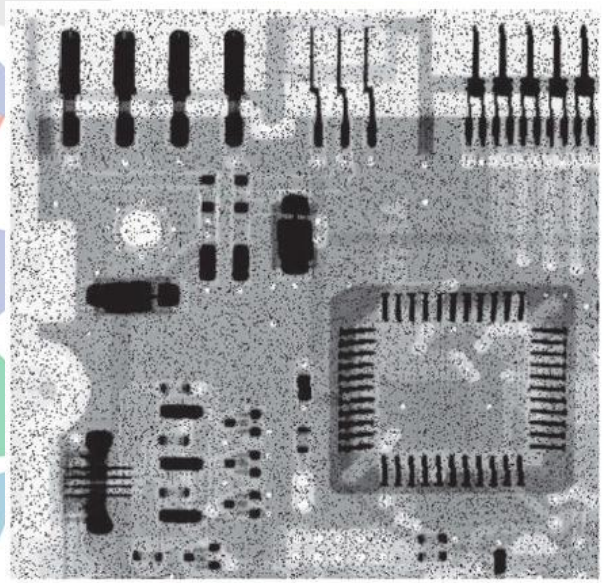
Q is the order of the filter.

Positive Q for pepper noise

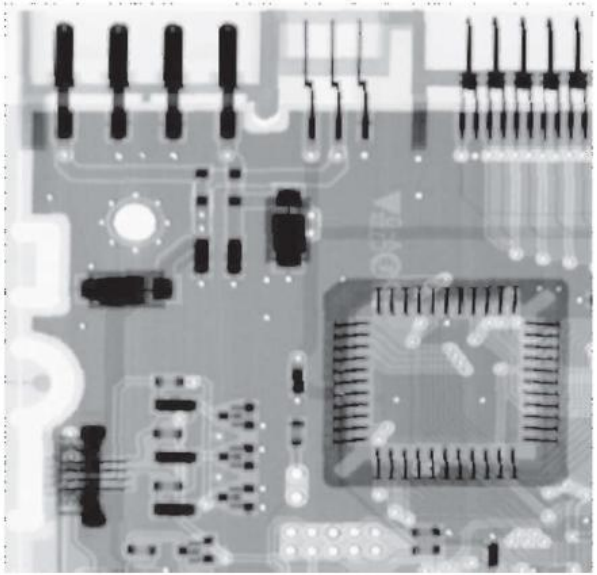
Negative Q for salt noise

Cannot do both simultaneously

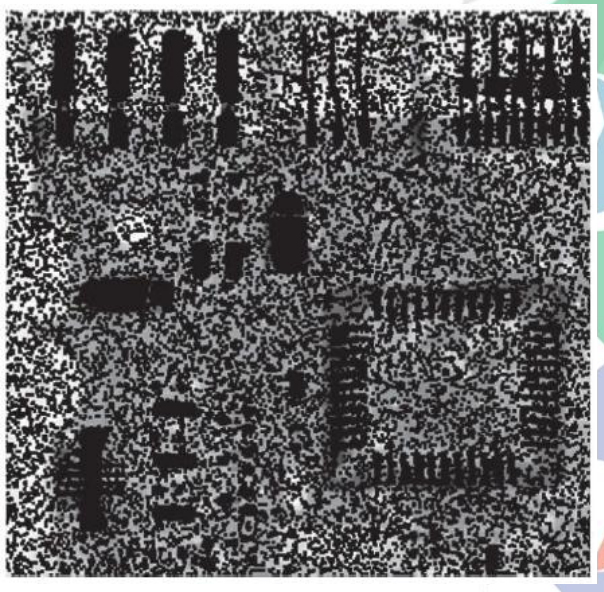
$$\hat{f}(x,y) = \frac{\sum_{(r,c) \in S_{xy}} g(r,c)^{Q+1}}{\sum_{(r,c) \in S_{xy}} g(r,c)^{Q}}$$



Pepper Noise
With p = 0.1

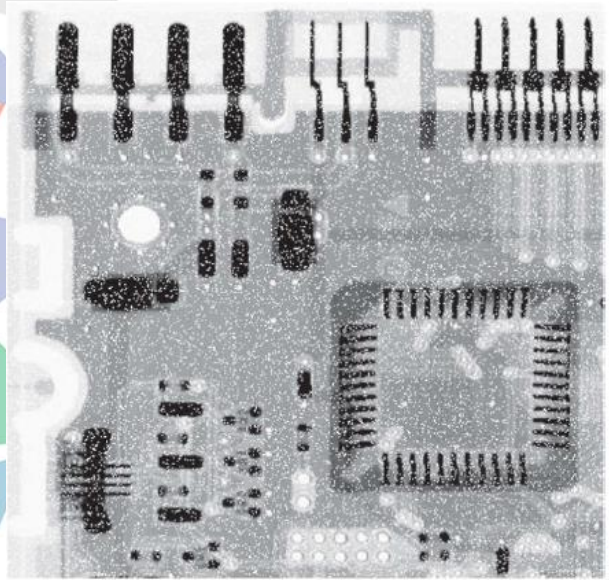


Filtered By
Contraharmonic filter
Q = 1.5

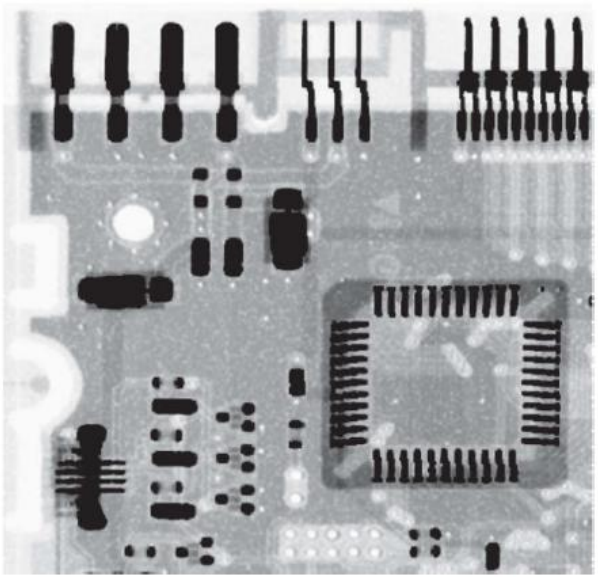


Filtered By
Contraharmonic filter
Q = -1.5

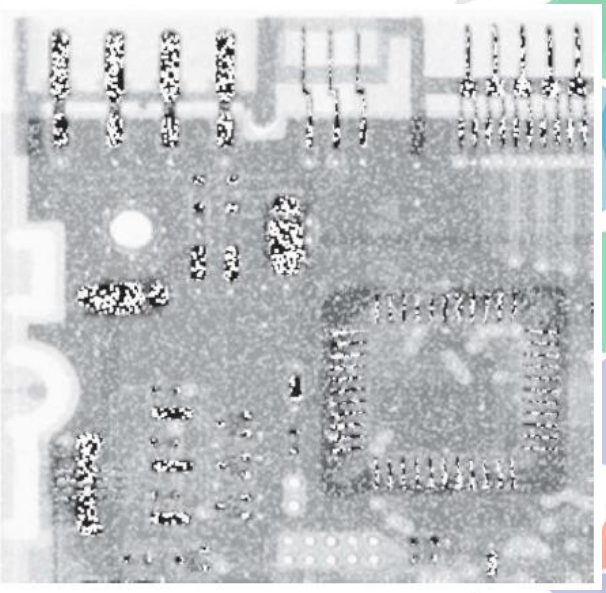
Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.



Salt Noise
With p = 0.1



Filtered By
Contraharmonic filter
Q = -1.5

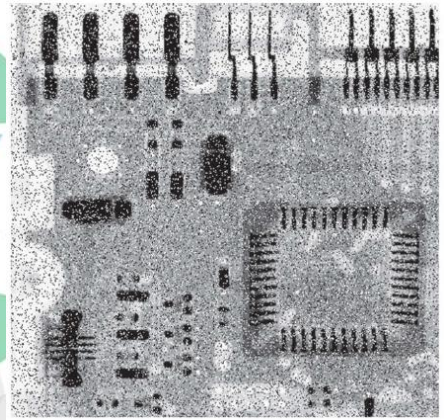


Filtered By
Contraharmonic filter
Q = 1.5

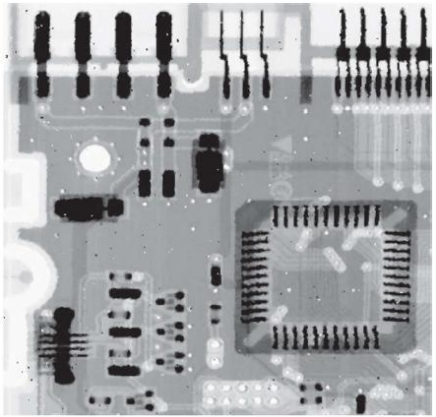
Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Order-Statistic Filters - Median

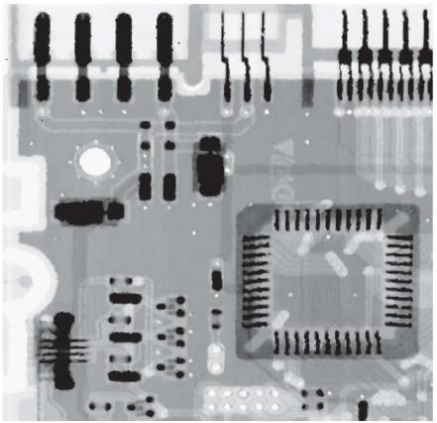
Simplest Order-Statistic Filter $\hat{f}(x,y) = \underset{(r,c) \in S_{xy}}{\text{median}} \{g(r,c)\}$



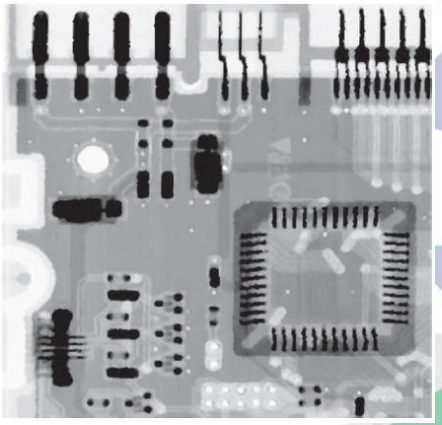
Salt and pepper Noise
With $p_s = p_p = 0.1$



1st pass
3x3 Median filter



2nd pass
3x3 Median filter



3rd pass
3x3 Median filter

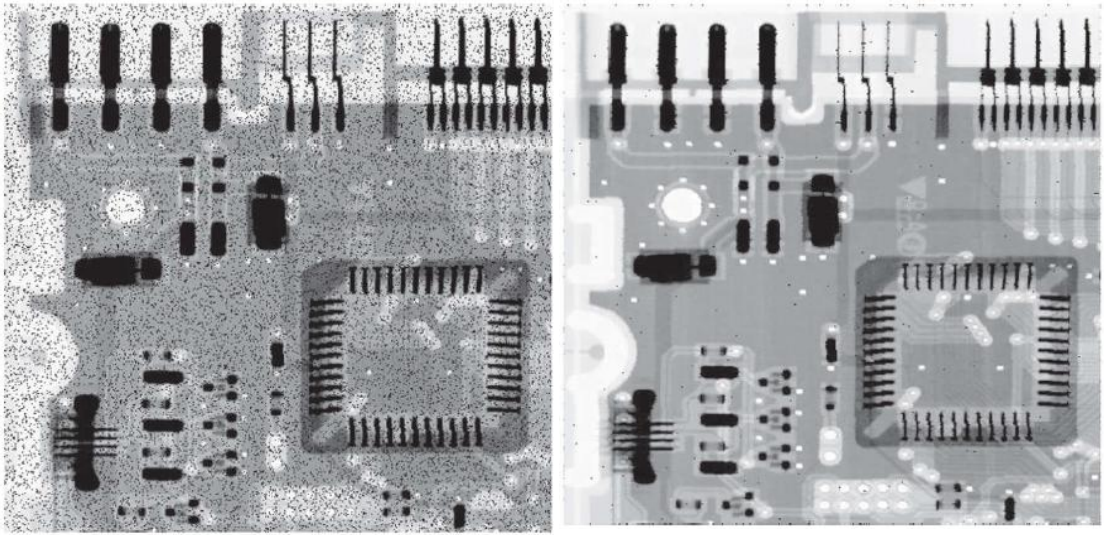
Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Order-Statistic Filters – Max and Min

Max is for pepper noise

$$\hat{f}(x,y) = \max_{(r,c) \in S_{xy}} \{g(r,c)\}$$

Pepper Noise With p = 0.1



Filtered by 3x3 Max filter

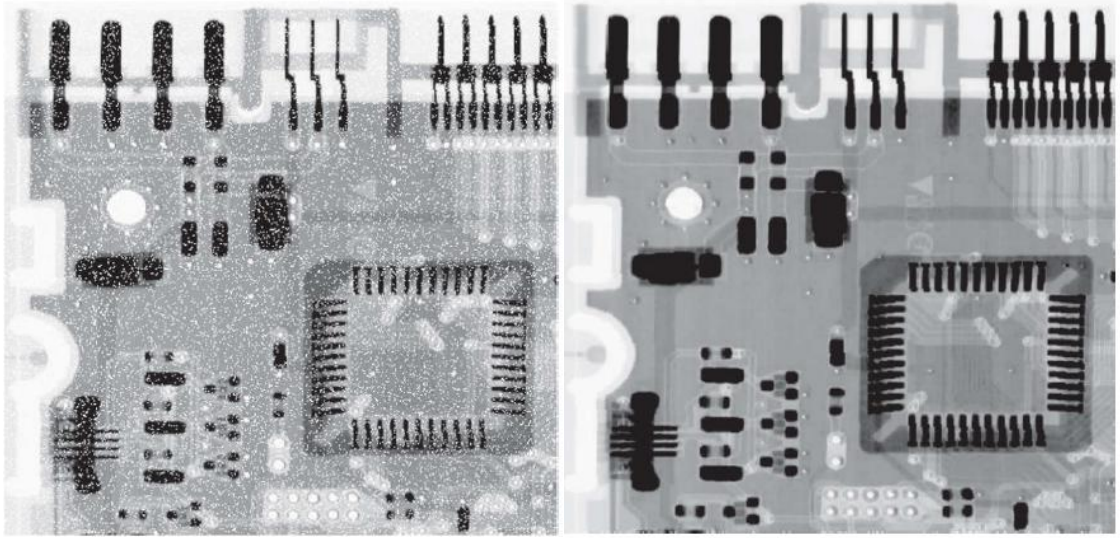
Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Order-Statistic Filters – Max and Min

Min is for salt noise

$$\hat{f}(x,y) = \min_{(r,c) \in S_{xy}} \{g(r,c)\}$$

Salt Noise With p = 0.1



Filtered by 3x3 Min filter

Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Order-Statistic Filters – Midpoint

Works well for randomly distributed noise – Gaussian, Uniform

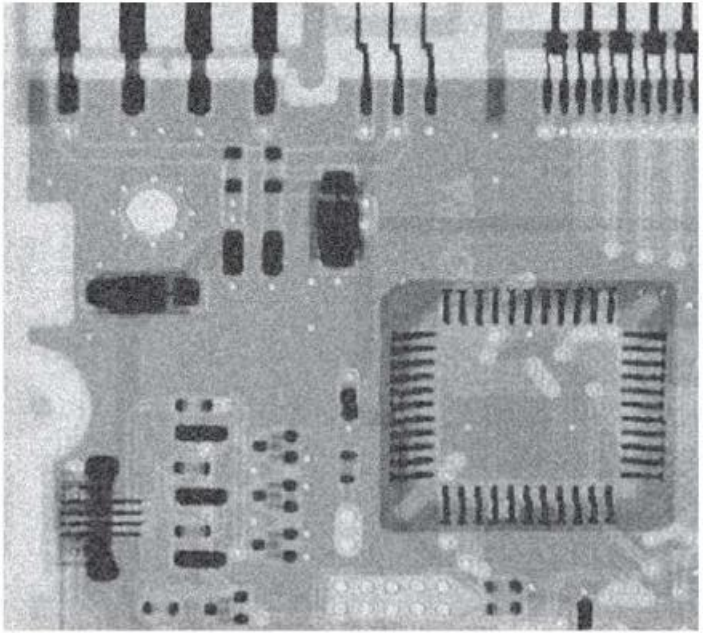
$$\hat{f}(x,y) = \frac{1}{2}\left[\max_{(r,c)\in S_{xy}}\{g(r,c)\} + \min_{(r,c)\in S_{xy}}\{g(r,c)\}\right]$$

Order-Statistic Filters – Alpha-Trimmed Mean

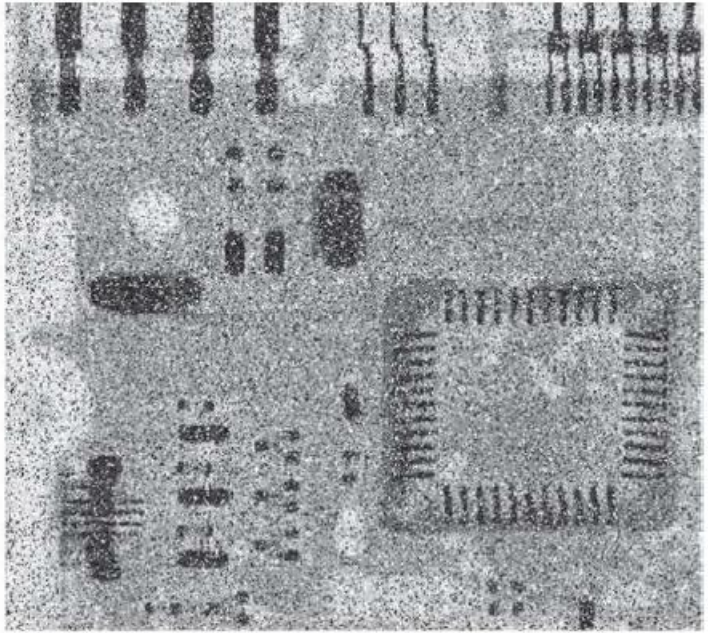
d is the number of values which will be deleted (d/2 lowest and d/2 highest intensity values of g(r,c)). After that, the remaining values will be averaged

$$\hat{f}(x,y) = \frac{1}{mn-d} \sum_{(r,c) \in S_{xy}} g_R(r,c)$$

Good for situations where there are multiple combined types of noise



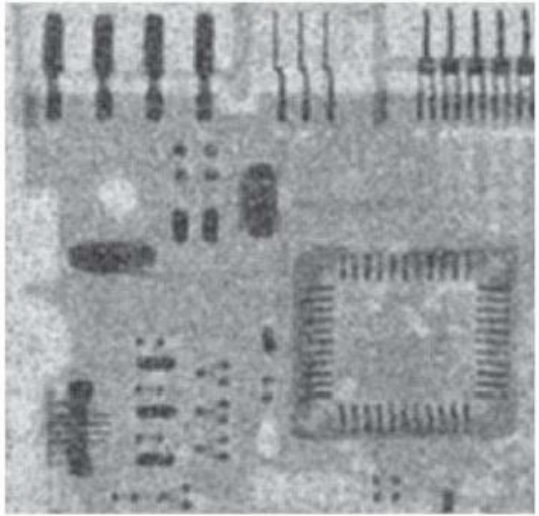
Uniform noise added

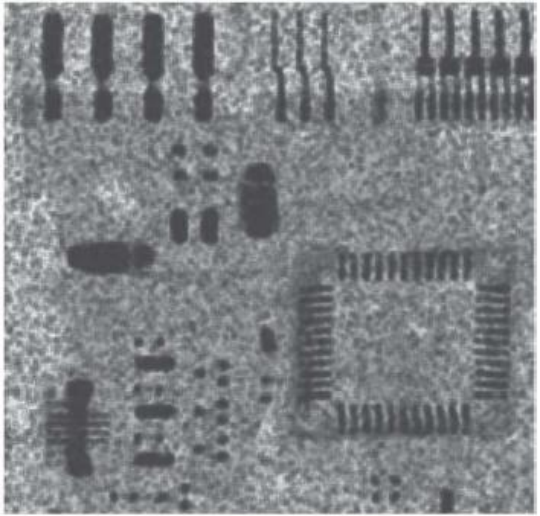


Salt-and-pepper noise added

Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

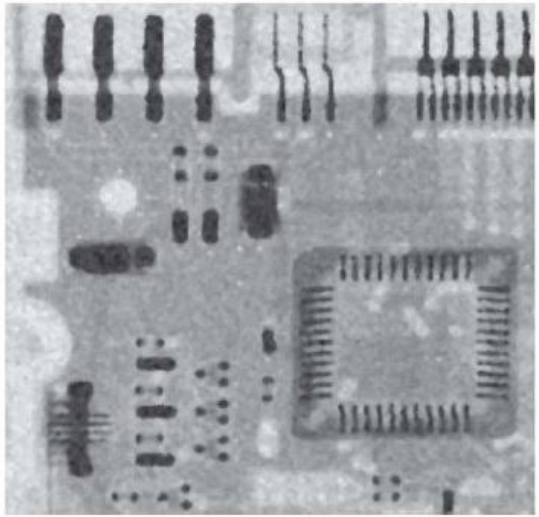
Arithmetic mean filter

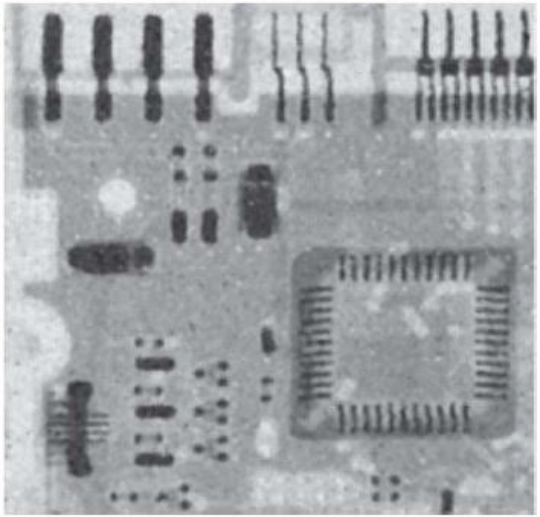




Geometric mean filter

Median filter





Alpha-trimmed Mean filter

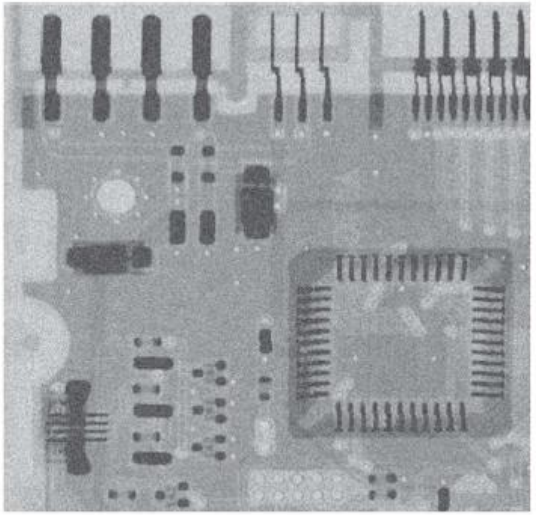
Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

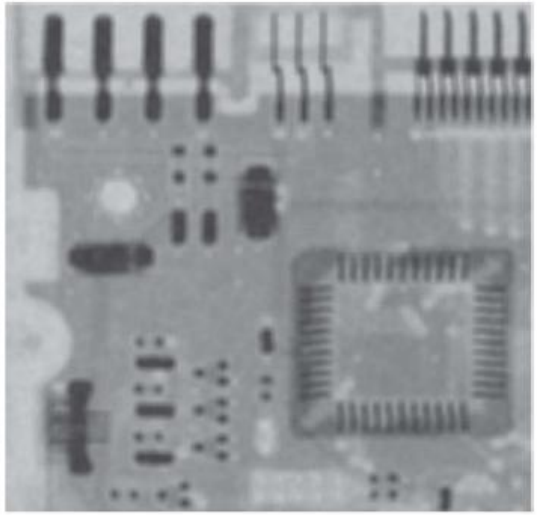
Adaptive Filter

Superior in result, inferior in computing cost

$$\hat{f}(x,y) = g(x,y) - \frac{\sigma_\eta^2}{\sigma_{S_{xy}}^2}[g(x,y) - \overline{z}_{S_{xy}}]$$

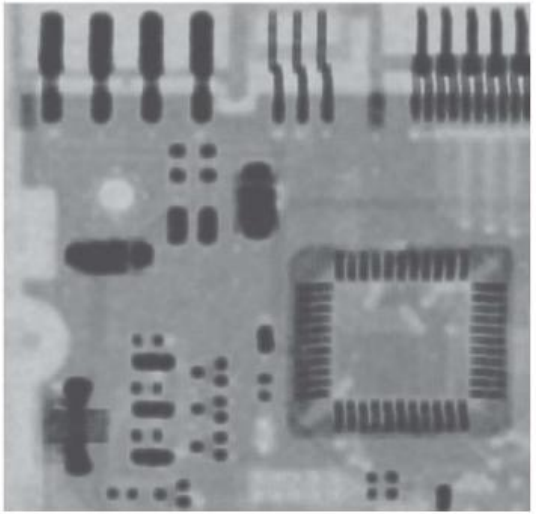
Corrupted by Gaussian noise

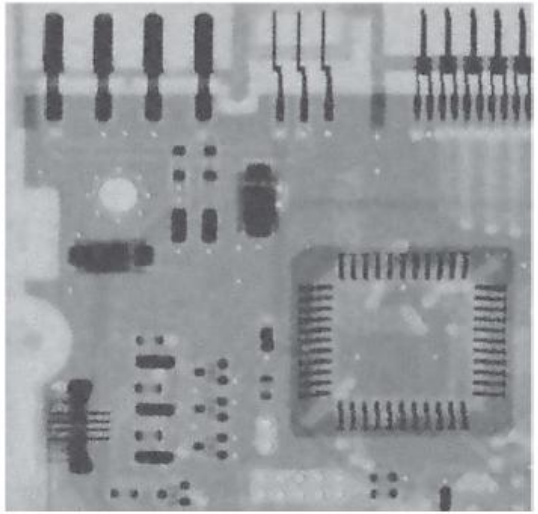




Arithmetic mean filter

Geometric mean filter





Adaptive filter

Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Noise Reduction – Frequency Domain Filtering

Notch Filter

$$H_{\mathrm{NR}}(u,v) = \prod_{k=1}^{Q} H_k(u,v) H_{-k}(u,v)$$

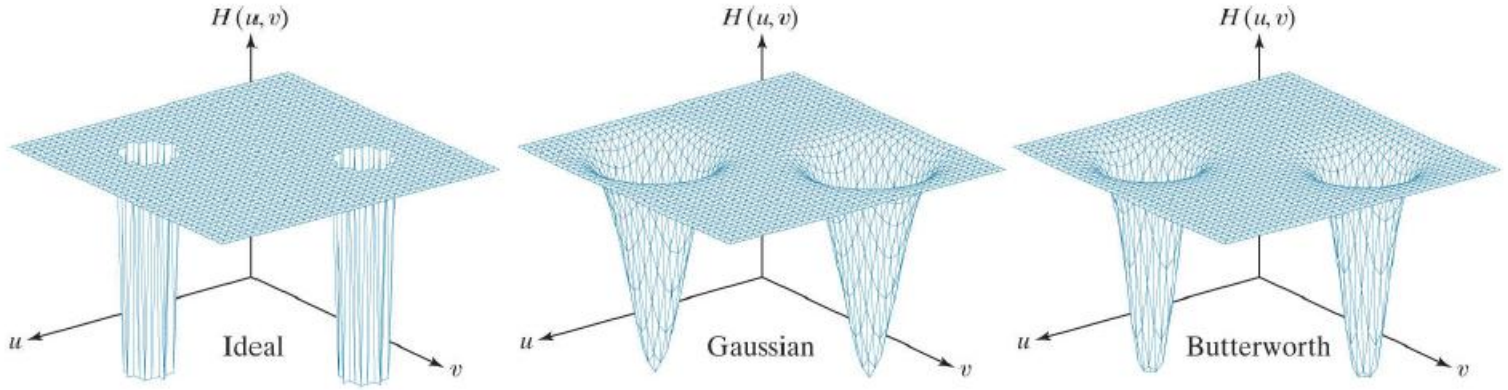
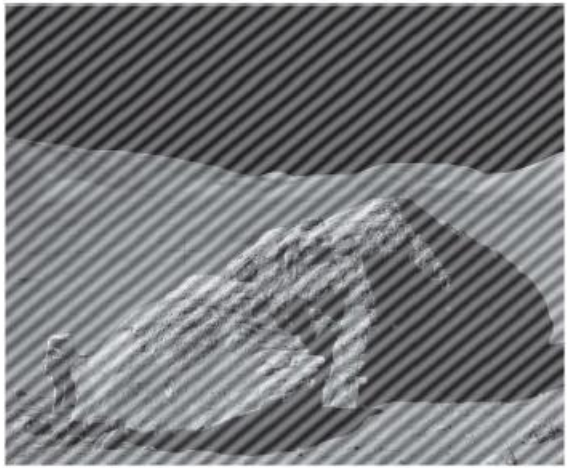
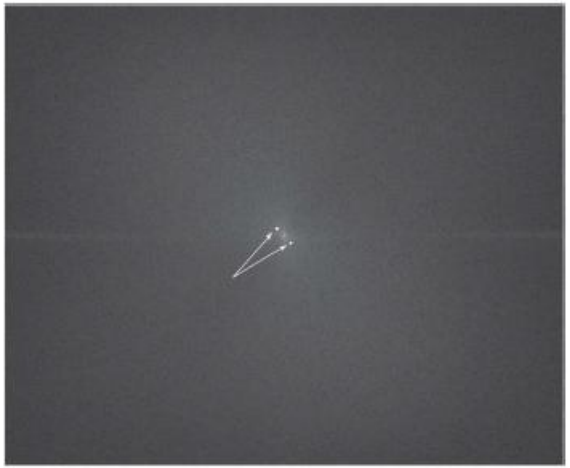


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Corrupted by sinusoidal noise





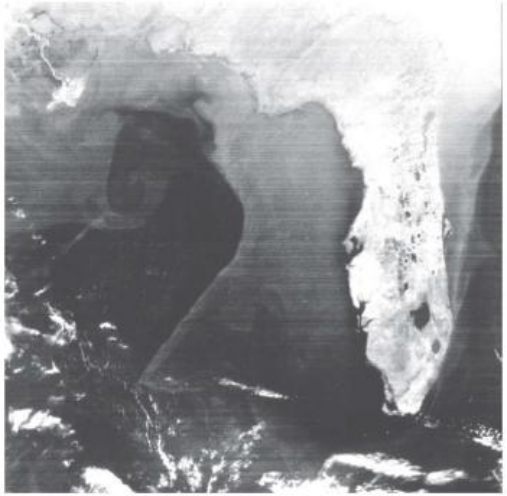
Spectrum in frequency domain

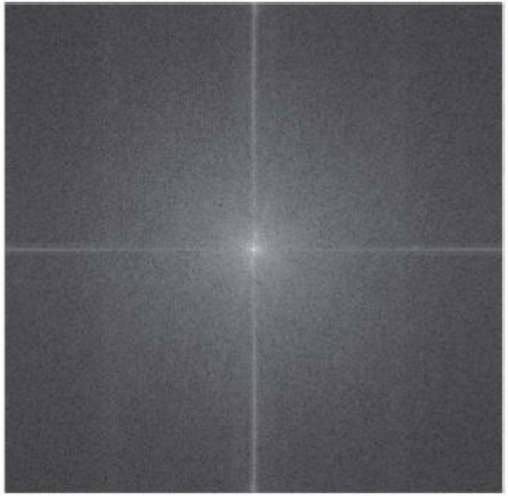
Notch filter

After filtered

Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

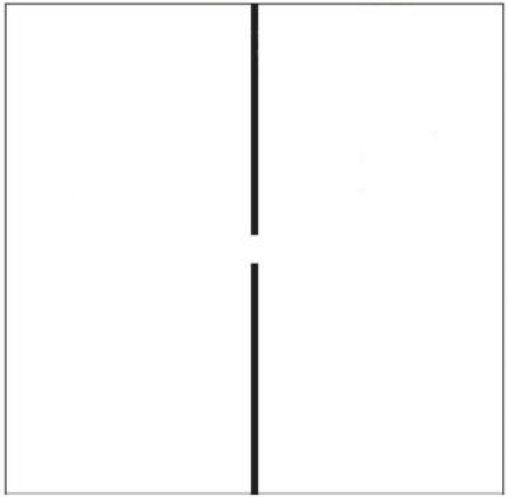
Corrupted by sinusoidal noise

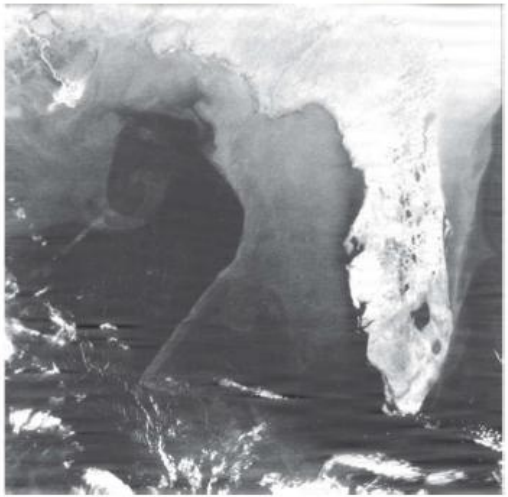




Spectrum in frequency domain

Notch filter



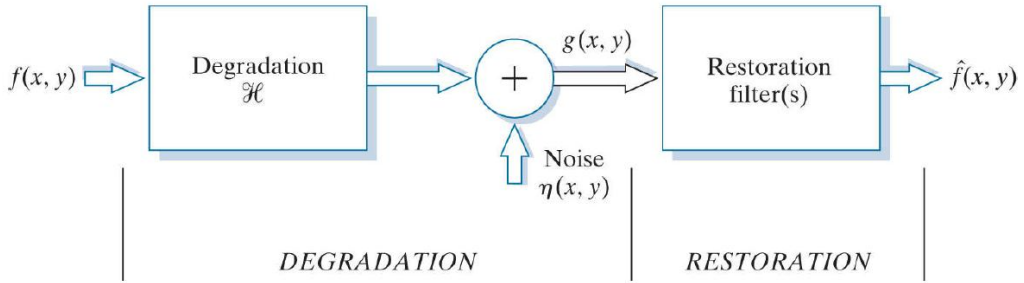


After filtered

Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Restore Degradation Function

Degradation function



- Back to this chart, the degradation function is h(x, y) which produces g(x, y) by operating on the image f(x, y)

$$g(x,y) = (h \star f)(x,y) + \eta(x,y)$$

$$G(u,v) = H(u,v)F(u,v) + N(u,v)$$

- Therefore, in order to restore the image, this function need to be inversed

Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Estimating the degradation function

The problem is, normally we do not know the degradation function. So, we need to estimate

There are 3 principle ways to estimate the degradation function

- **Observation** – observe through a small section of the image
- **Experimentation** – If the degradation is occurred by an equipment, it might be possible to reproduce the degradation by experimenting various settings on the equipment
- **Modeling**

Inverse Filtering

$$G(u,v) = H(u,v)F(u,v) + N(u,v)$$

- As the forward operation in frequency domain is as above, the inverse operation is

$$\hat{F}(u,v) = F(u,v) + \frac{N(u,v)}{H(u,v)}$$

- Therefore, the operation is to convert the image into Fourier Series -> inverse the function -> convert back to spatial domain

Example

$h(x,y) = \frac{1}{2\pi\sigma^2} e^{-r^2/2\sigma^2}$

$n(x,y)$ = Normal distribution, mean zero

$f(x,y)$

$g(x,y)$

blur $\sigma = 1.0$ pixels

noise $\sigma = 0.3$ grey levels

Image from: A. Zisserman (2014). B14 Image Analysis, University of Oxford

noise σ = 0.3 grey levels

$\hat{F}(u,v) = G(u,v) / H(u,v)$

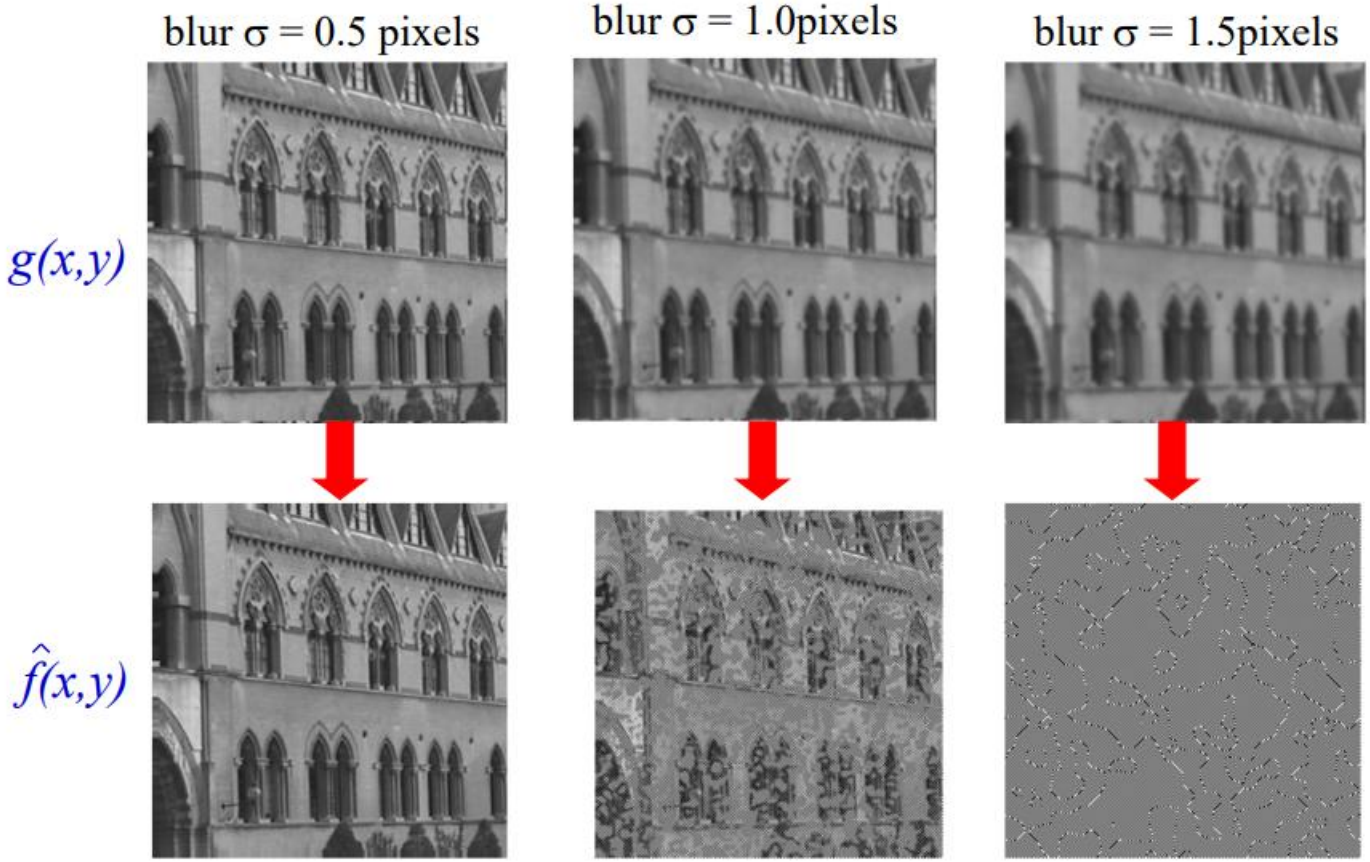


Image from: A. Zisserman (2014). B14 Image Analysis, University of Oxford

Inverse Filtering

$$\hat{F}(u,v) = F(u,v) + \frac{N(u,v)}{H(u,v)}$$

- As you can see that if H(u, v) has zero or small values, the second term will dominate the whole image – As we can see in the case in the last slide

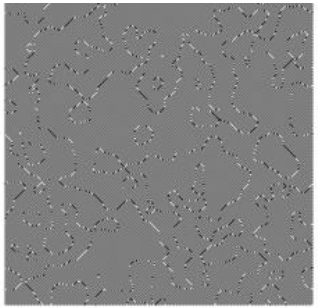


Image from: A. Zisserman (2014). B14 Image Analysis, University of Oxford

Inverse Filtering

- This is the problem of noise amplification

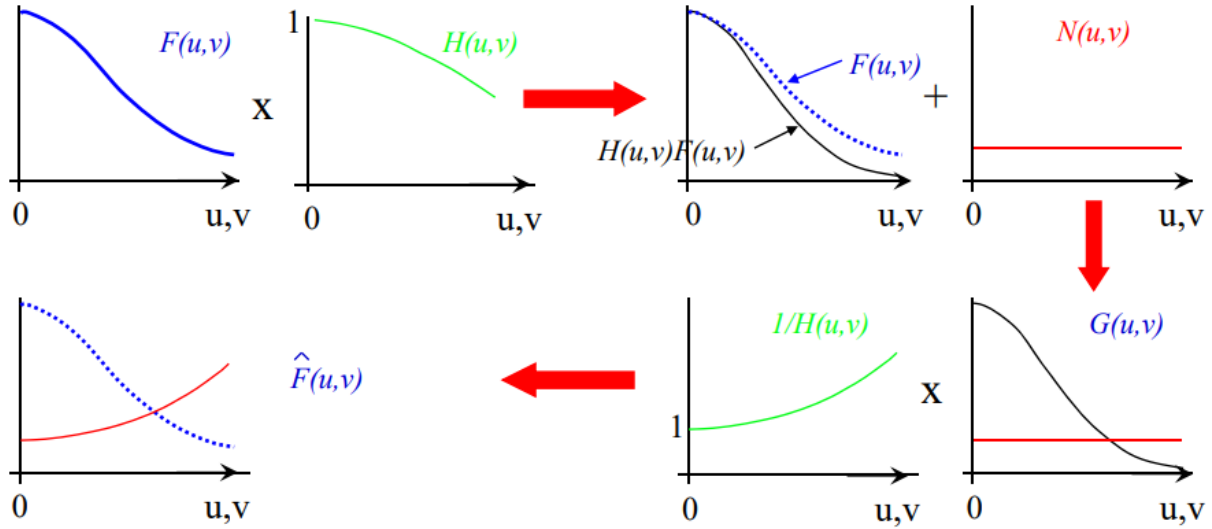


Image from: A. Zisserman (2014). B14 Image Analysis, University of Oxford

The Wiener Filtering

- Minimum Mean Square Error (Wiener) filter is a restoration filtering which can handle noise

$$\hat{F}(u,v) = W(u,v)\, G(u,v)$$

$$W(u,v) = \frac{H^*(u,v)}{|H(u,v)|^2 + K(u,v)}$$

- Where the least square error of the image is $\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [f(x,y) - \hat{f}(x,y)]^2$

Image from: A. Zisserman (2014). B14 Image Analysis, University of Oxford

The Wiener Filtering

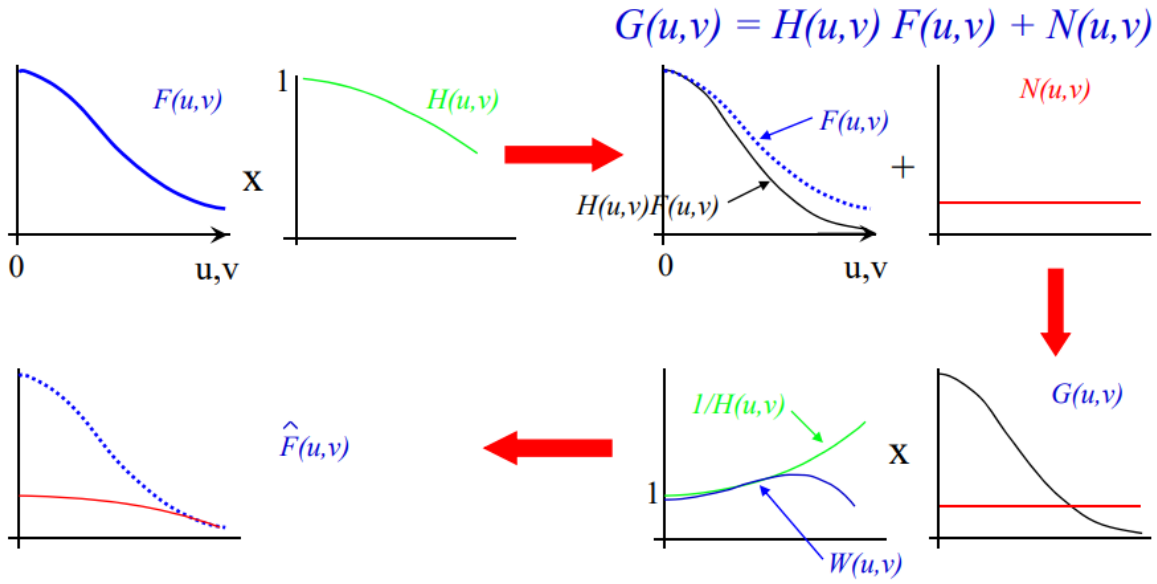


Image from: A. Zisserman (2014). B14 Image Analysis, University of Oxford

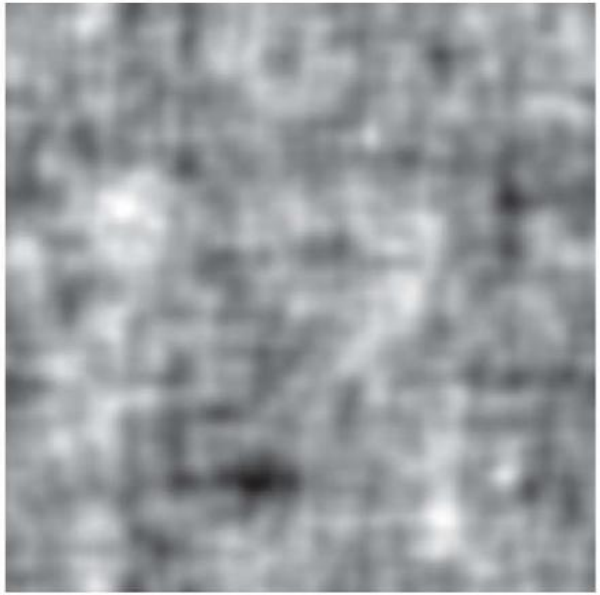
The Wiener Filtering

Original Image

Degraded image

Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

The Wiener Filtering



Full inverse filtering

Wiener filter

Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

The Wiener Filtering

blur σ = 1.5 pixels
noise σ = 0.3 grey levels

$\hat{F}(u,v) = W(u,v)\, G(u,v)$

$$W(u,v) = \frac{H^*(u,v)}{|H(u,v)|^2 + K(u,v)}$$

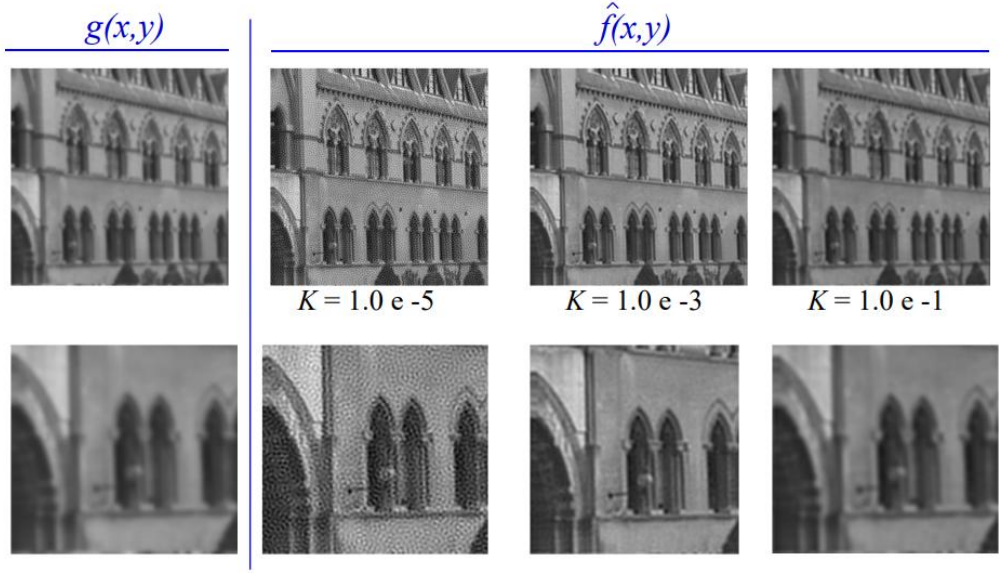


Image from: A. Zisserman (2014). B14 Image Analysis, University of Oxford

Case study 1

Degradation model

$$g(x,y) = \frac{1}{T}\int_{-T/2}^{T/2} f(x - x_0(t), y)\, dt$$

Require $H(u,v)$ for Wiener filter

Image from: A. Zisserman (2014). B14 Image Analysis, University of Oxford

$$G(u,v) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)e^{-j2\pi(ux+vy)}dxdy,$$

$$= \frac{1}{T}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\left\{\int_{-T/2}^{T/2} f(x-x_0(t),y)dt\right\}e^{-j2\pi(ux+vy)}\,dxdy$$

interchange order of spatial and temporal integration

$$G(u,v) = \frac{1}{T}\int_{-T/2}^{T/2}\left\{\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x-x_0(t),y)e^{-j2\pi(ux+vy)}\,dxdy\right\}\,dt$$

Fourier transform of $f(x-x_0(t),y)$.

$$G(u,v) = \frac{1}{T}\int_{-T/2}^{T/2} F(u,v)e^{-j2\pi ux_0(t)}\,dt$$

$$= F(u,v)\frac{1}{T}\int_{-T/2}^{T/2} e^{-j2\pi ux_0(t)}\,dt$$

$$= F(u,v)H(u,v)$$

where

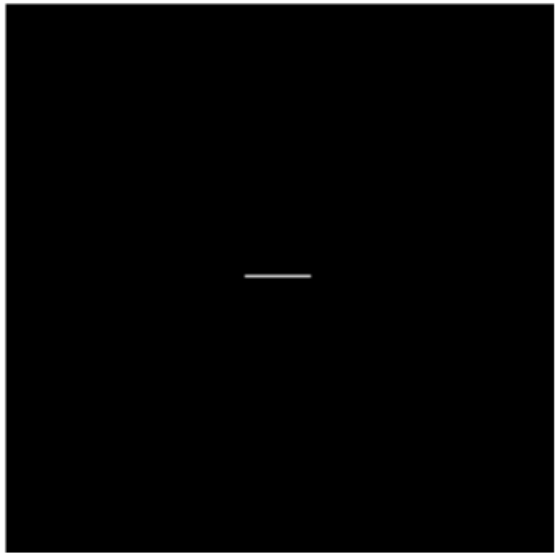
$$H(u,v) = \frac{1}{T}\int_{-T/2}^{T/2} e^{-j2\pi ux_0(t)}\,dt$$

Image from: A. Zisserman (2014). B14 Image Analysis, University of Oxford

suppose $x_0(t) = st$, and $sT = d$ pixels

FT of … $h(x, y) = \text{hat}_d(x)\delta(y)$

$$
\begin{aligned}
H(u,v) &= \frac{1}{T}\int_{-T/2}^{T/2} e^{-j2\pi ust}\, dt \\
&= \frac{1}{T}\left[\frac{e^{-j2\pi ust}}{-j2\pi us}\right]_{-T/2}^{T/2} \\
&= \frac{1}{j2\pi ud}\left(e^{j\pi ud} - e^{-j\pi ud}\right) \\
&= \text{sinc}\pi ud
\end{aligned}
$$



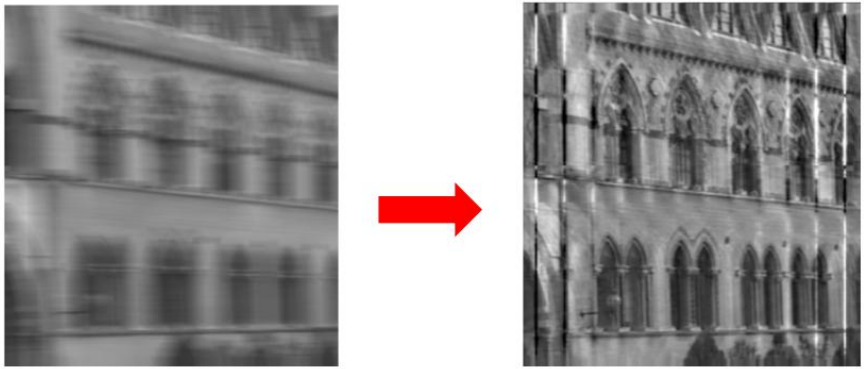
Note, *H(u,v)* has zeros – a problem for an inverse filter

Image from: A. Zisserman (2014). B14 Image Analysis, University of Oxford

Case study 1

blur = 20 pixels

$$W(u,v) = \frac{H^*(u,v)}{|H(u,v)|^2 + K(u,v)}$$



1. Compute the FT of the blurred image
2. Multiply the FT by the Wiener filter $\hat{F}(u,v) = W(u,v)\, G(u,v)$
3. Compute the inverse FT

Image from: A. Zisserman (2014). B14 Image Analysis, University of Oxford

Case study 2

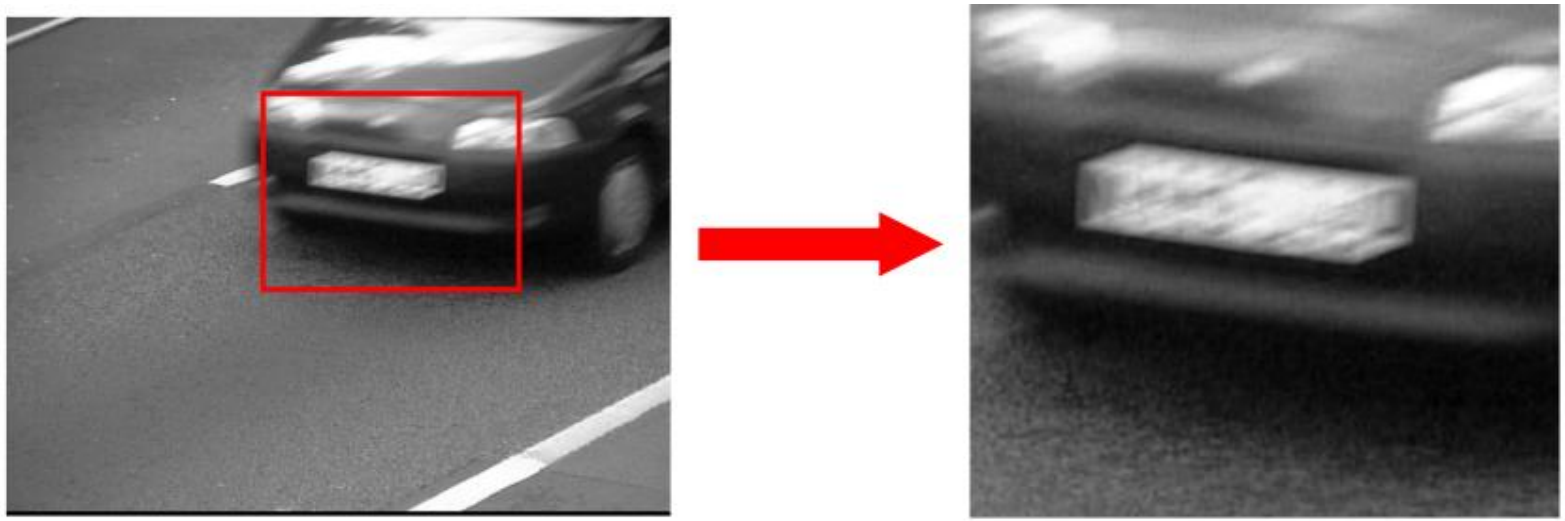


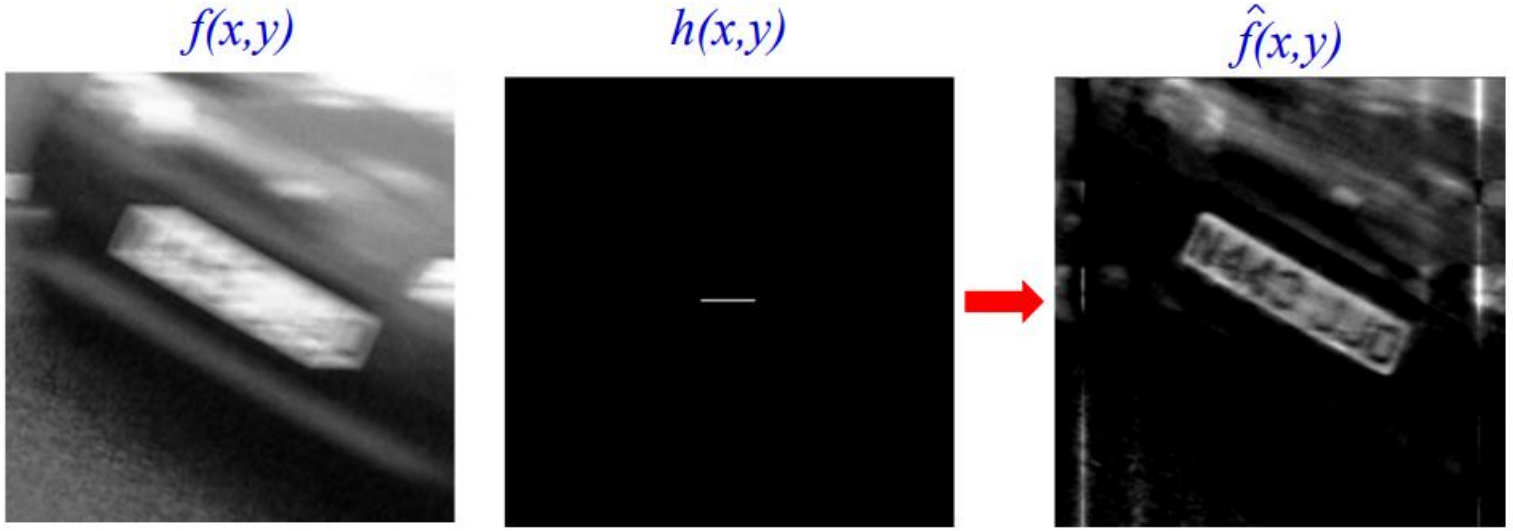
Image from: A. Zisserman (2014). B14 Image Analysis, University of Oxford

Algorithm

1. Rotate image so that blur is horizontal
2. Estimate length of blur
3. Construct a bar modelling the convolution
4. Compute and apply a Wiener filter
5. Optimize over values of K

Image from: A. Zisserman (2014). B14 Image Analysis, University of Oxford

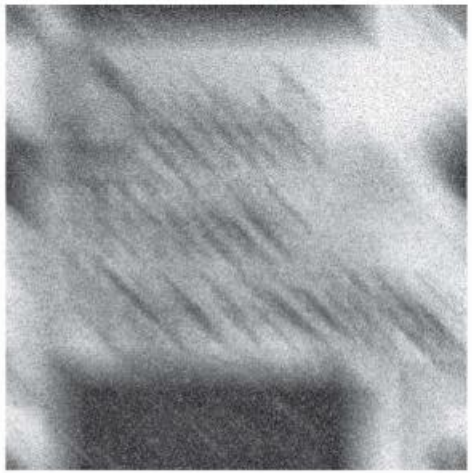
Case study 2



blur = 30 pixels

Image from: A. Zisserman (2014). B14 Image Analysis, University of Oxford

Case study 3



Degraded image

Full inverse filtering

Wiener filter

Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Case study 3

Degraded image
With noise variance one order of magnitude less

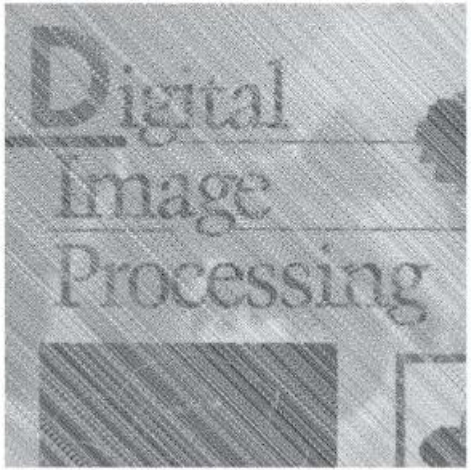
Full inverse filtering

Wiener filter

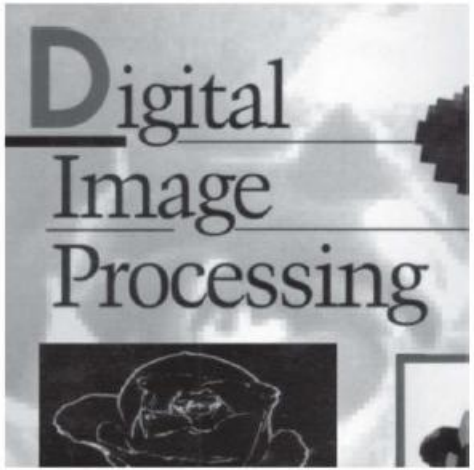
Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Case study 3

Degraded image
With noise variance five orders of magnitude less



Full inverse filtering



Wiener filter

Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

End of Week 6